

ECE501: Digital Image Processing

Project Topic: Adaptive Fabric Surface Defect Detection and Localization using Classical Texture Descriptors

GROUP 19

Week 2 Progress Report

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Deliverable: Literature study of Local Binary Patterns (LBP) and design of the LBP-based detection stage

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Objectives of the Week

- Study published work on LBP-based fabric defect detection in depth [1], [2].
- Understand the uniform, rotation-invariant LBP operator and its (P, R) scales.
- Decide how LBP histograms will be computed on defect-free reference patches and on test patches.
- Explain why LBP is the first descriptor we study, and prepare the comparison with GLCM and Gabor features planned for the coming weeks.

Work Done This Week

A. Literature study

In Week 1 we noted that classical descriptors such as LBP and Gabor filters have been used for fabric inspection but are often not adaptive, while deep detectors such as the lightweight YOLOv5s variant of Kang [3] are fast and accurate (about 102 FPS) but need labelled training data and more computing power. This week we looked at how LBP can be made adaptive without any labels.

Tajeripour et al. [1] compute LBP histograms on windows of a defect-free reference image and use a threshold to flag defective windows. The method works on plain and patterned fabrics and is simple enough for online use. In their tests, the detection rate was above 95% for all defect types, using 16 x 16 detection windows on 256 x 256 images and only five defect-free images for training. They also found that combining two scales, LBP with 8 points on a 3 x 3 window and 16 points on a 5 x 5 window, worked

better than a single scale. This is the closest published match to our project, because the normal appearance is learned only from defect-free fabric and the threshold is set from that reference.

Our plan differs from [1] in three ways. Their method uses square windows, a log-likelihood ratio, and the maximum score of defect-free windows as the threshold. We plan to use the circular LBP operator of Ojala et al. [2], the chi-square distance, and a threshold of the form mean + k times standard deviation.

We also noted the contrast with GLCM-based methods, which uses co-occurrence statistics and will be studied next week, and with Gabor filter banks, which depend on orientation and frequency.

B. The LBP operator

For each pixel, LBP compares the P neighbours on a circle of radius R with the centre pixel and forms a binary code [2]:

$$\text{LBP}_{\{P,R\}} = \sum_{i=0}^{P-1} s(g_i - g_c) 2^i$$

Here g_c is the centre gray value, g_i is the i-th neighbour, and $s(x)$ is 1 if x is at least 0 and 0 otherwise.

A code is called uniform if it has at most two transitions between 0 and 1 around the circle. Uniform codes are grouped by the number of 1s, and all non-uniform codes share one extra bin. The rotation-invariant uniform histogram therefore has $P + 2$ bins: 10 bins at $(P = 8, R = 1)$ and 18 bins at $(P = 16, R = 2)$. Each patch is described by this histogram, normalised to sum to 1.

C. Why LBP first

- LBP only compares each neighbour with the centre pixel, so the code does not change when gray levels change monotonically [2]. Uneven lighting is one of our main challenges, because it can be mistaken for a defect.
- The feature vector is short (10 or 18 values). This suits our situation of having few defect-free samples per fabric type, and [1] also trained from only five images.
- LBP needs one pass of pixel comparisons and a histogram, with no co-occurrence matrix or filter bank. This fits our aim of a lightweight alternative to deep detectors [3].
- The reference-histogram, window and threshold scheme of [1] maps almost directly onto our adaptive design.
- Only defect-free fabric is used to build the reference, so no defect annotation is needed.
- LBP (a local micro-pattern descriptor) and GLCM (a second-order statistics descriptor) are different in nature, which makes them a meaningful pair for the planned comparison.

Descriptor	Strength	Concern
LBP [2]	Cheap, compact histogram, invariant to monotonic gray-level changes	Sensitive to noise in flat regions; may miss defects that change only brightness
GLCM	Rich statistics such as contrast and homogeneity	Needs quantization and several distances and orientations
Gabor	Captures oriented structure	A filter bank costs more and has more parameters to tune

D. LBP stage design

- Pre-processing: grayscale conversion and Gaussian or median denoising. Denoising matters for LBP because small noise differences in flat regions can flip the neighbour comparisons. CLAHE will be tested as an optional step.
- Patch extraction: each image is divided into fixed-size patches (starting at 128 x 128 pixels, to be tuned), following the window-based analysis of [1].
- LBP features [2]: rotation-invariant uniform LBP at ($P = 8$, $R = 1$) and ($P = 16$, $R = 2$). The two scales can be used alone or concatenated.
- Normal-texture reference: the mean histogram of defect-free patches, one model per fabric type, in the spirit of [1]. A covariance model would need regularisation, because LBP histograms sum to 1 and their covariance matrix is singular.
- Anomaly score: the distance between a test-patch histogram and the reference. We start with the chi-square distance. It is suited to comparing histograms because it gives more weight to differences in bins with small values than plain Euclidean distance does. Mahalanobis scoring will be revisited when the descriptors are combined.
- Adaptive threshold: $\tau = \text{mean} + k \times \text{standard deviation}$, computed from scores of defect-free patches (k to be tuned between 2 and 3).
- Post-processing: morphological opening and closing, and removal of very small regions, to turn flagged patches into a localization mask.

E. Evaluation plan (preview)

Patch-level detection will be scored with accuracy, precision, recall and F1, and pixel-level localization with IoU and Dice. These metrics will be defined formally in Week 3, together with the dataset choice, so that LBP and GLCM can be compared on the same footing. We are considering public fabric databases with pixel-level masks; the final choice will be made in Week 3.

Literature Review

Topic / Paper	Key Takeaway	Relevance to Our Project
Tajeripour et al., Fabric Defect Detection Using Modified Local Binary Patterns [1]	Builds an LBP reference from defect-free fabric, compares window histograms against it and uses a threshold to find defective windows. Works on plain and patterned fabrics; detection rate above 95%; trained from five defect-free images; combining two scales works best.	Closest to our approach: a normal-texture reference, window analysis and a threshold. Main basis of the LBP stage.
Ojala et al., Multiresolution gray-scale and rotation invariant	Introduces uniform, rotation-invariant LBP at several (P , R) scales.	Source of our LBP operator and of the two scales (8, 1) and (16, 2).

Topic / Paper	Key Takeaway	Relevance to Our Project
texture classification with LBP [2]		
Kang, Research on fabric defect detection method based on lightweight network [3]	Improves YOLOv5s with attention and feature-enhancement modules and a CloU loss; about 102 FPS.	Shows what deep detectors achieve and what they cost; motivates our lightweight, adaptive classical pipeline.

Results and Analysis

This week was devoted to study and design, so the LBP stage has not yet been run on the full data. Quantitative results (patch accuracy, precision, recall, F1, pixel IoU and Dice) will be reported in Week 4 using the metrics above. Based on the literature, we expect LBP to detect defects that change the local texture, such as holes, knots and broken threads [1]. Because LBP is unchanged by monotonic gray-level shifts [2], a stain that changes brightness but not texture may be harder to detect on grayscale images. These are expectations to be tested, not findings.

Challenges Faced

- Noise sensitivity: LBP codes in flat regions can change with small amounts of noise, so the denoising step and its strength need to be tuned.
- Choice of (P, R): a small radius captures fine weave structure, while a larger radius captures coarser patterns. We plan to compare the two scales and their combination.
- Patch size: small windows localize defects better but give unstable histograms; large windows do the opposite.
- Patterned fabrics: normal texture can vary across the image, so a single mean histogram may not be enough for such fabrics.
- Dataset choice: public databases differ in fabric type, defect classes and availability of pixel-level masks and defect-free images.
- Threshold factor k: it must balance false alarms against missed defects.

Plan for Next Week (Week 3)

- Study published work on GLCM in the same depth as LBP.
- Shortlist public fabric defect datasets with defect-free and defective images.
- Extend the LBP design into the complete pipeline: pre-processing, feature extraction, normal-texture reference, anomaly scoring, thresholding and post-processing.
- Define how detection and localization quality will be measured.

Conclusion

In Week 2 we studied LBP-based fabric defect detection and designed the LBP stage of our pipeline: rotation-invariant uniform LBP histograms on patches, a normal-texture reference built from defect-free fabric, and an adaptive threshold. We chose LBP first because it is robust to lighting changes, compact,

cheap to compute, and directly supported by the reference-and-threshold scheme of [1], while deep models are accurate but heavier [3]. Week 3 will widen the study to GLCM and Gabor features and fix the datasets and evaluation plan.

References

- [1] F. Tajeripour, E. Kabir, and A. Sheikhi, "Fabric defect detection using modified local binary patterns," *EURASIP J. Adv. Signal Process.*, vol. 2008, Art. no. 783898, 2008. doi: 10.1155/2008/783898.
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- [3] X. Kang, "Research on fabric defect detection method based on lightweight network," *J. Eng. Fibers Fabr.*, vol. 19, 2024. doi: 10.1177/15589250241232153.